

Chapter 6.A

Thermochemistry

Chemical Principles I

CHEM 1211K

1. Define and apply work and heat.
2. Describe and apply the concept of conservation of energy.
3. Define and apply state and path functions or variables.
4. Apply the concept of state functions to predict the energy changes involved in reactions.
5. Calculate pressure-volume work.
6. Calculate the heat required to change the temperature of a substance.
7. Summarize the roles of enthalpy and heat flow in chemical reactions.
8. Calculate changes in enthalpy and internal energy in chemical reactions.
9. Apply Hess's law to calculate the enthalpy of a target reaction given known enthalpies of related reactions.
10. Apply enthalpies of formation and Hess's law to calculate the enthalpy of a target reaction.

- **Thermochemistry** is the study of heat and energy in chemical reactions.
- **Energy**
 - **Mechanical energy** is due to the motion or position of an object.
 - **Kinetic energy** is energy of motion.
 - **Potential energy** is energy related to position.

- The SI unit of work, heat, and energy is the **Joule (J)**. One Joule is equivalent to $1 \text{ kg}\cdot\text{m}^2/\text{s}^2$.
 - 1 liter-atmosphere is equal to 101.325 Joules.
 - 1 calorie is equal to 4.184 Joules.
 - 1 Calorie (nutritional calorie) is equal to 1000 calories or 4184 Joules.

- Proper definitions of the system, surroundings, and universe are key to solving problems in thermodynamics.
- **System**
- **Surroundings**
- **Universe**

- **Open system:** both matter and energy can be exchanged between the system and the surroundings
- **Closed system:** energy—but *not* matter—can be exchanged between the system and the surroundings
- **Isolated system:** neither energy nor matter can be exchanged between the system and the surroundings



Figure 6.3. Examples of open, closed, and isolated systems.

- **Work** is the result of a force acting through a distance—a kind of energy transfer in which the particles of the system undergo *coordinated motion* in a nonrandom direction.

- **Heat** is the result of a transfer of thermal energy—a kind of energy transfer in which the particles of the system undergo *random motion*.

- In chemistry, the sign of work is defined with reference to energy transfer to or from the system.
 - Work done **on** the system by the surroundings is **positive**.
 - Work done **by** the system on the surroundings is **negative**.

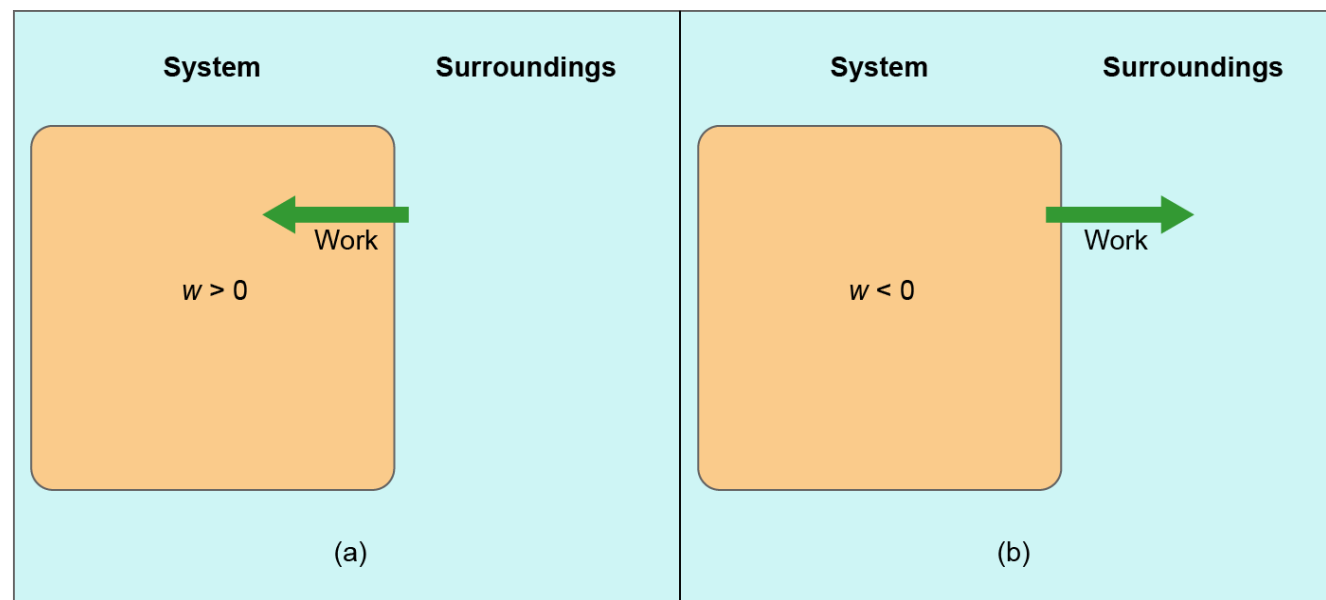


Figure 6.4. Work done on the system is positive; work done by the system is negative.

- The sign of heat is also defined with reference to energy transfer to or from the system.
 - Heat (thermal energy) transferred **to** the system from the surroundings is **positive**.
 - Heat (thermal energy) transferred **from** the system to the surroundings is **negative**.

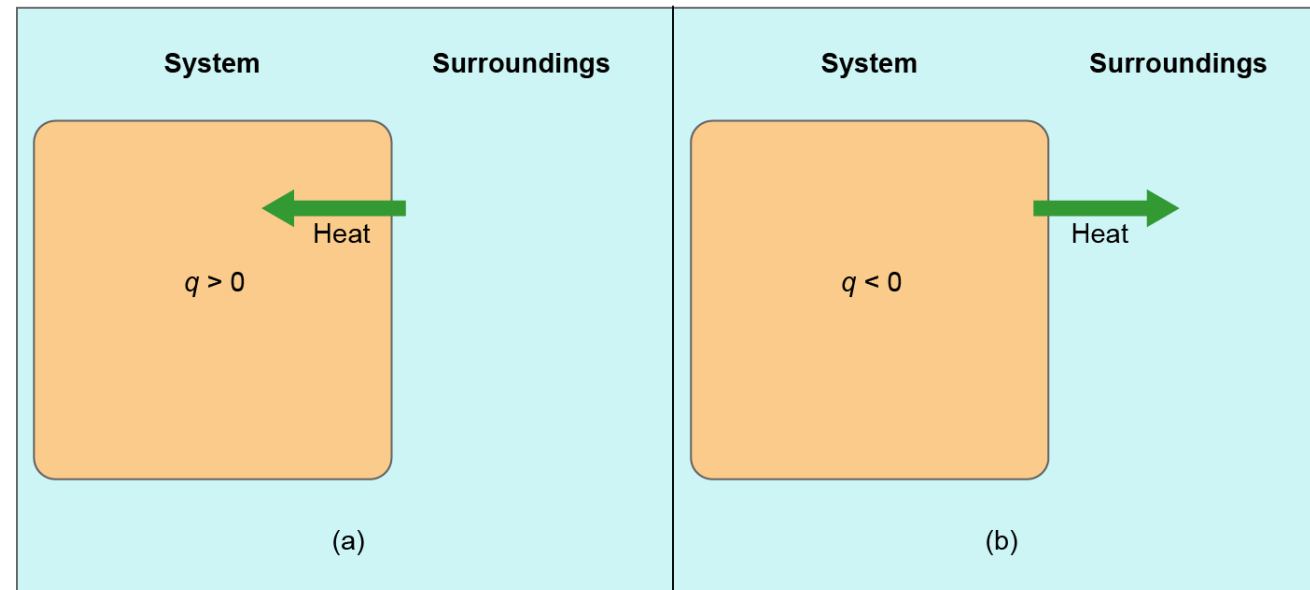


Figure 6.5. Heat transferred to the system is positive; heat transferred from the system is negative.

Is work done on the system or by the system in the following scenarios? In each case, the system is bolded.

1. A **person** rows a boat across a lake.
 - A. Work is done on the system ($w > 0$).
 - B. Work is done by the system ($w < 0$).

2. A person pedals a **bicycle** down the street.
 - A. Work is done on the system ($w > 0$).
 - B. Work is done by the system ($w < 0$).

- **Internal energy (U or E)** is the sum of the kinetic and potential energies of all particles within the system.
- **The first law of thermodynamics**
 - The total energy of the universe is constant.
 - Energy is neither created nor destroyed.
 - All energy transfers (changes in internal energy) are due to heat and/or work.
- A change in internal energy is denoted ΔU ...

- All changes in internal energy ΔU are due to heat and/or work. We can express this idea mathematically and will frequently use this equation as a starting point when solving problems in thermochemistry.
- Recall the sign conventions for internal energy, heat, and work.

Parameter	Sign	Meaning
q	+	Heat flows into the system.
q	–	Heat flows out of the system.
w	+	Work is done on the system.
w	–	Work is done by the system.
ΔU	+	The system gains internal energy.
ΔU	–	The system loses internal energy.

Table 6.2

For a particular process $q = -10$ kJ and $w = +25$ kJ. Which statement is *true*?

- A. Heat flows from the surroundings to the system.
- B. The system does work on the surroundings.
- C. ΔU is equal to -35 kJ.
- D. All these statements are true.
- E. None of these statements is true.

6.3 Energy as a State Function | 1 of 3

- **State variables (functions)** have values that depend only on the internal state of the system—the positions and velocities of all particles.
 - Well defined for an unchanging thermodynamic system
 - Changes in state variables are easy to measure and calculate: $\Delta X = X_2 - X_1$.
 - A “state function” is just some mathematical expression that includes only state variables.
- Internal energy is one example of a state variable.
- You may already be familiar with many thermodynamic state variables: volume, mass, temperature, number of moles, pressure, ...
- We will encounter other important state variables later in this chapter and in chapter 18.

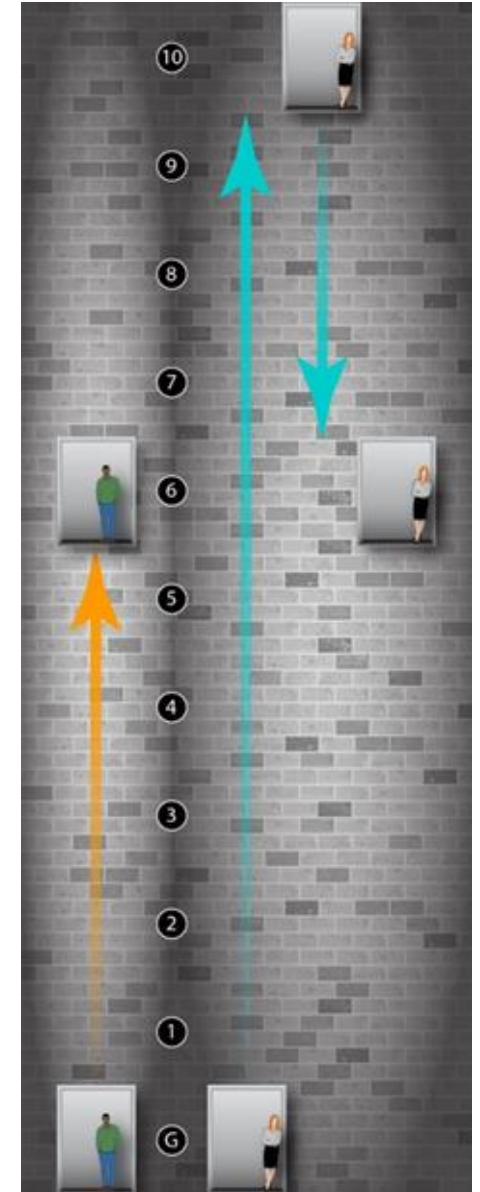


Figure 6.7. Elevator height H is a state variable.

- Work and heat are **path functions (path variables)**, quantities whose values depend on the path or *process* taken between two states, the initial and final states.

- Consider 1 mole of liquid ethanol at 20 °C and atmospheric pressure.
 - What do we know about this sample from just this information? The **state variables**: P , T , n , U , etc.
 - The ethanol could have reached this state in different ways; each is associated with different values for the **path variables** q and w . For example,
 - A. Solid ethanol at –120 °C is heated directly to 20 °C.
 - B. Solid ethanol at –120 °C is heated first to 50 °C and then cooled to 20 °C.
 - The change in internal energy (ΔU) is the same in both cases, but q and w differ for the two different paths.

- Work is about more than just pushing macroscopic objects against friction or gravity.
- The work done on a gas to compress it (or by the gas as it expands) is called **pressure-volume (PV) work**.
- PV work can be expressed in units based on the product of a pressure and volume, such as **liter-atmospheres ($L \cdot atm$)**.

The change in internal energy for a given system is -325 J . This system loses 155 J of energy as heat as it expands from an initial volume of 15.0 L at 1.00 atm of pressure. What is the final volume of the system in liters?

One liter-atmosphere is equivalent to 101.3 Joules ; one Joule is equivalent to $0.00987\text{ liter-atmospheres}$.

- **Heat and temperature are not the same.** The amount of heat required to raise the temperature of 1 gram of a substance by 1 degree Celsius is called the **specific heat (c)** of the substance.

$$q = m \times c \times \Delta T$$

- Note that these equations only hold in the absence of a phase change. We will address phase changes later in the course.

- The **molar heat capacity** (C_m) of a substance is the heat required to raise the temperature of 1 mole of the substance by 1 degree Celsius.

$$q = n \times C_m \times \Delta T$$

- Specific heat and molar heat capacity are intensive properties; **heat capacity** (C) itself is extensive.

$$q = C \times \Delta T$$

Substance	Specific Heat ($\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$)	Substance	Specific Heat ($\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$)
CO_2	0.852	Al	0.892
CO	1.04	Cr	0.45
H_2	14.4	Co	0.46
N_2	1.04	Cu	0.385
O_2	0.922	Au	0.129
$\text{H}_2\text{O(g)}$	2.042	Fe	0.442
$\text{H}_2\text{O(l)}$	4.184	Pb	0.13
$\text{H}_2\text{O(s)}$	2.089	Mg	1.0
		Ag	0.24
		Na	0.293
		Sn	0.22
		Zn	0.388

Table 6.3. Specific heats of select substances.

What is the specific heat capacity of a substance if 1155 J of heat are needed to raise the temperature of 100. g of the substance from 25.0 °C to 55.0 °C? Enter your response in Joules per degree Celsius per gram to the nearest 0.001 J °C⁻¹ g⁻¹.

Chapter 6.B

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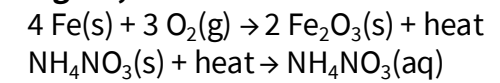
- Most chemical reactions are run in open systems in which both heat and work can be nonzero. Usually, the **pressure is constant** at or near atmospheric pressure.
- We are typically more interested in the heat (q) than the work (w) of chemical reactions.
- Starting from the first law, we can identify a new state function **enthalpy** ($H = U + PV$), the change in which is equal to the heat of reaction at constant pressure.

$$\Delta U = q + w$$

- **Exothermic** processes
- **Endothermic** processes

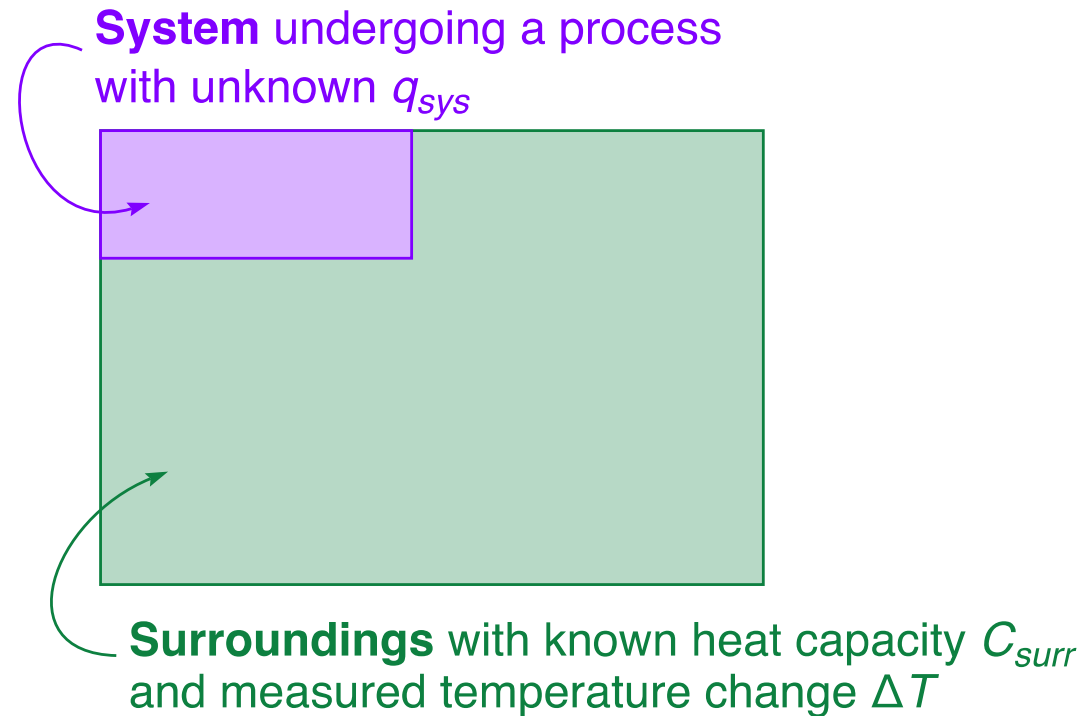


Figure, section 6.4.



- **Calorimetry** is a laboratory technique for quantifying heat transfer by measuring temperature changes in the substances involved.
- Calorimetry calculations are based on a **heat balance** of the system and surroundings:

$$q_{sys} = -q_{surr} = -C_{surr}\Delta T$$



A 25.0 g block of metal initially at 95.0 °C is placed in a calorimeter containing 121 g of water initially at 27.2 °C. The specific heat of the metal is 0.49 J °C⁻¹ g⁻¹. The specific heat of water is 4.184 J °C⁻¹ g⁻¹.

Assuming that no heat is lost to the calorimeter or surroundings, what is the final temperature of the water and metal? Enter your response in degree Celsius to the nearest 0.1 °C. [Start by drawing a picture and writing a heat balance.](#)

6.6 Calorimetry: Measuring Energy Changes | 2 of 4

- The system of interest in chemical calorimetry is typically a submicroscopic chemical reaction. The heat evolved or absorbed by the system is determined by measuring the temperature change in the *surroundings*.

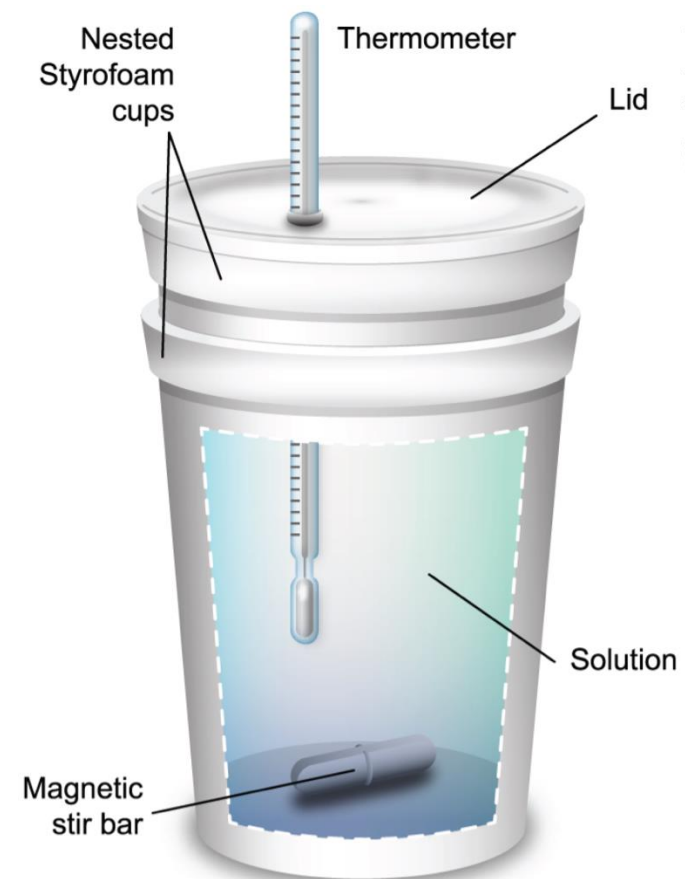


Figure 6.13. A coffee-cup calorimeter.

- **Constant-pressure calorimetry** involves running a reaction in an insulated vessel under atmospheric pressure. In this situation, the heat absorbed or released by the reaction (q_p) is equal to the enthalpy change (ΔH).
- The reaction is typically happening in aqueous solution.
 - The *system* is the submicroscopic reactant and product molecules.
 - The *surroundings* include the reaction solution itself and the calorimeter walls.

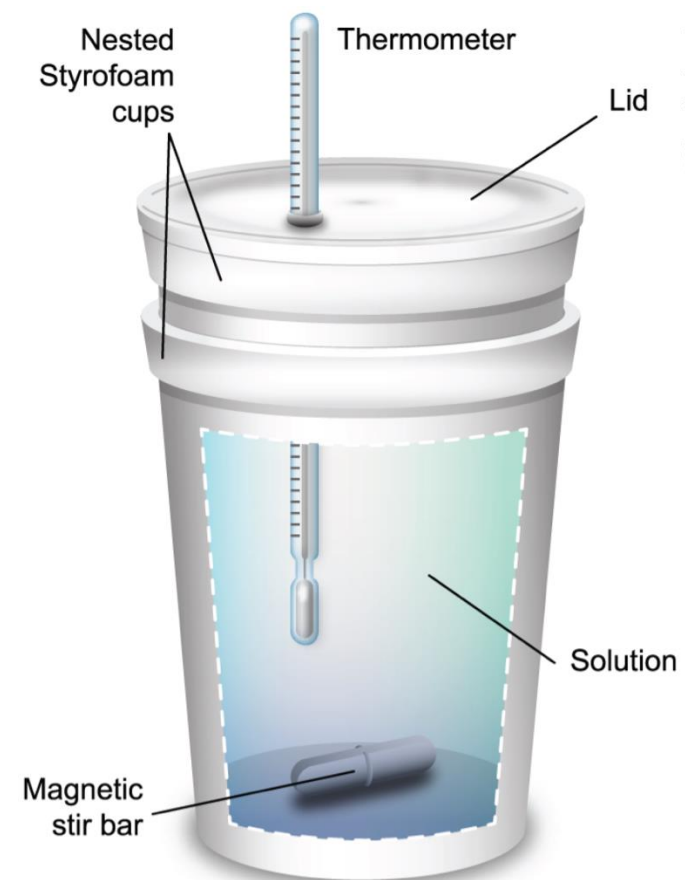
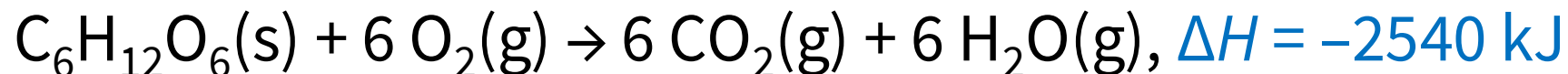


Figure 6.13. A coffee-cup calorimeter.

- Keep in mind that the measured solution is part of the surroundings and the system is considered closed. The measured ΔT reflects energy transfer to or from the surroundings—*NOT* the system of interest.
- **Increase** in solution **temperature**
 - Heat was released by the reaction to the surroundings and the reaction is **exothermic**.
- **Decrease** in solution **temperature**
 - Heat was absorbed by the reaction from the surroundings and the reaction is **endothermic**.

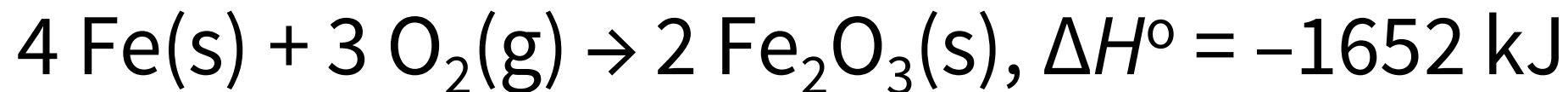
A 20.0 mL sample of 0.50 M NaOH and a 20.0 mL sample of 0.50 M HNO₃ are combined in a coffee-cup calorimeter. The temperature of the combined solutions increases from 20.4 °C to a maximum of 26.1 °C. Given that the heat capacity of the calorimeter is 25.2 J/°C, what is ΔH for the reaction in kJ/mol? You may assume that the density of the solution is 1.00 g/mL and the specific heat capacity is that of water (4.184 J·°C⁻¹·g⁻¹).

- **Thermochemical equation:** a balanced chemical equation that also includes the enthalpy change or **heat of reaction** (ΔH_{rxn}) for 1 mole of reaction events of the equation as written



- The heat of reaction can be used in molar ratios for stoichiometric calculations (“How much heat...”).
In other words, heat is now part of our stoichiometric toolbox!

The overall reaction in a commercial heat pack can be represented as follows.



What is the change in enthalpy when 2.00 g of Fe(s) is reacted with excess oxygen?

Enter your response in kilojoules per mole to the nearest 0.1 kJ/mol and with the appropriate sign.

Chapter 6.C

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- **Hess's law** is based on the idea that enthalpy is a state function.
 ΔH is the same regardless of the path followed to “get from” reactants to products.

- Per Hess's law, for a reaction that can be written as the sum of multiple steps, the enthalpy is equal to the sum of the enthalpies of the individual steps.

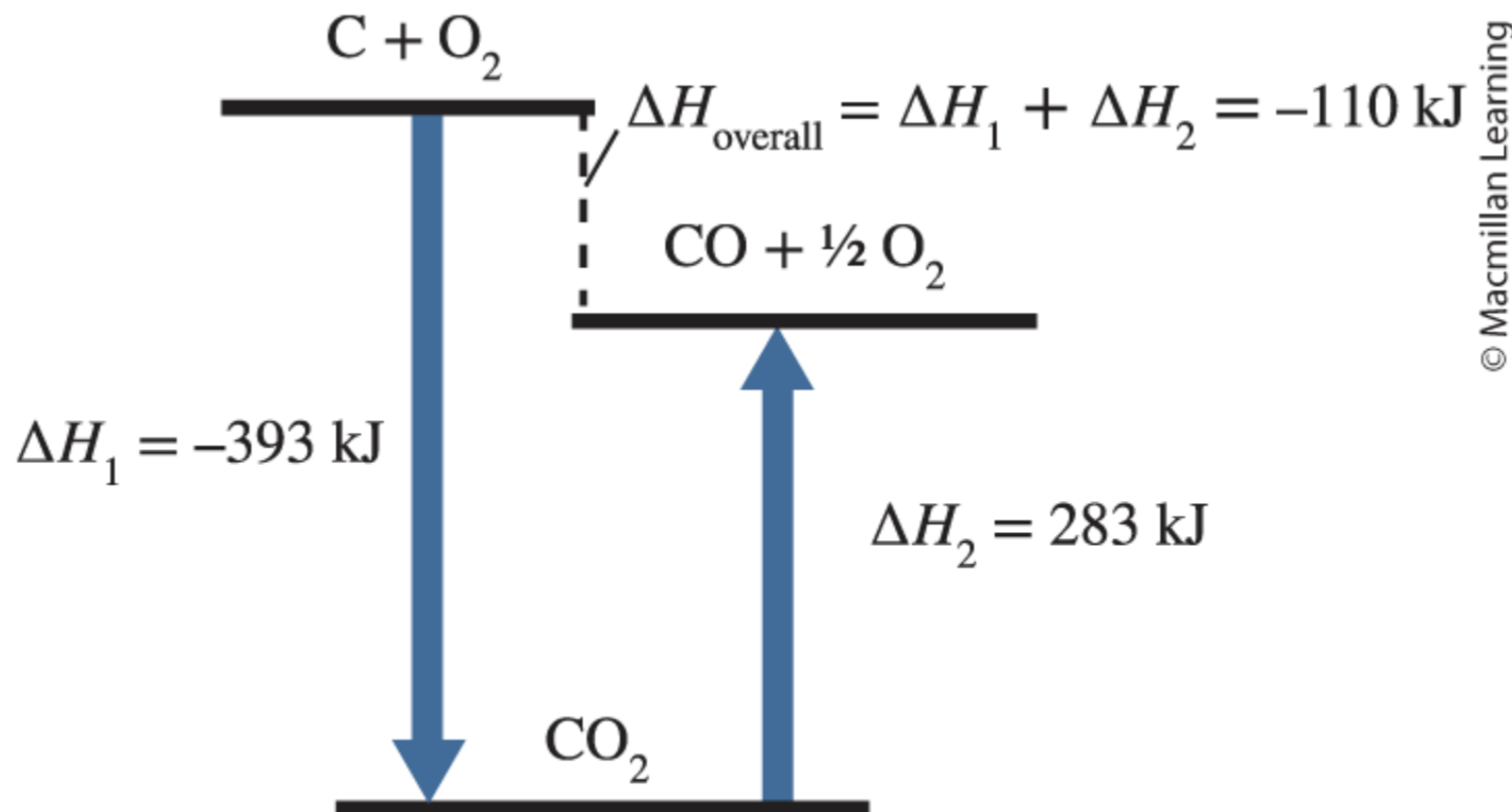
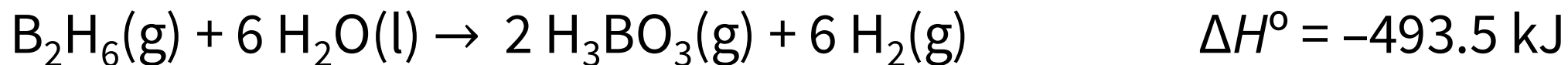
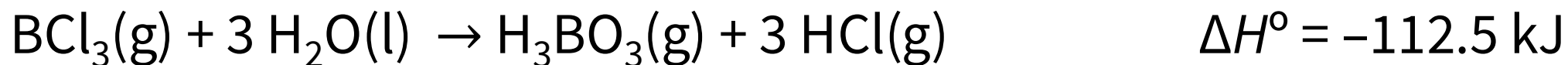
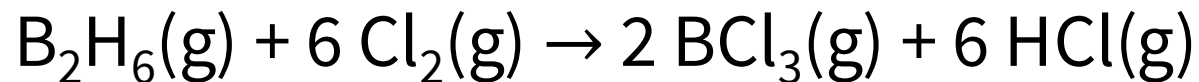


Figure 6.16. The enthalpy of reaction for oxidation of C to CO is $\Delta H_1 + \Delta H_2$.

- Per Hess's law, when a chemical equation is scaled or reversed, ΔH for the reaction is scaled or negated, respectively.
 - If you reverse a reaction, change the sign of ΔH .
 - If you multiply a reaction by n , also multiply ΔH by n .
 - If you add two reactions, add their ΔH values.

Use the information below to determine the standard enthalpy change for the reaction below in kilojoules (kJ).



- 41

- Because of the way formation is defined, ΔH_f° for an element in its standard state is zero.
- Standard enthalpies of formation are *measured quantities* that will be provided in questions where needed. They can also be found in the Appendix of your textbook or in a large list [here](#).

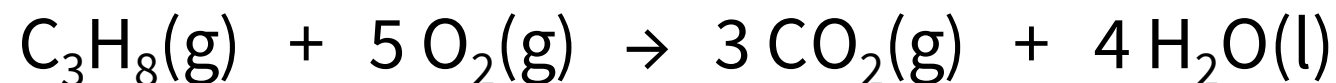
- Standard enthalpies of formation can be used to systematically calculate enthalpies of reaction.
- When the reaction involves multiple moles of reactants or products (nonzero coefficients)...

Example. Calculate the standard enthalpy of reaction (ΔH°_{rxn}) for the reaction below from the given data.

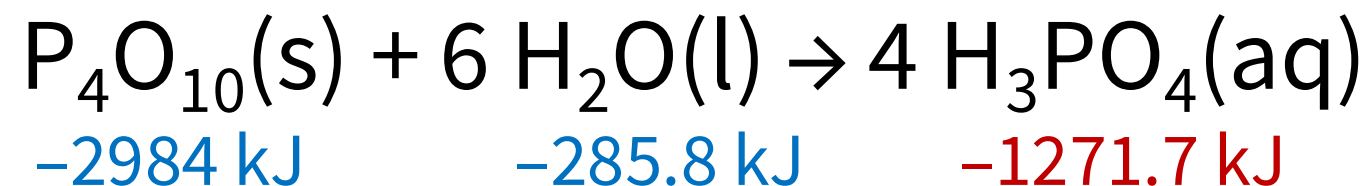
$$\Delta H_f^\circ[\text{C}_3\text{H}_8(\text{g})] = -103.8 \text{ kJ}$$

$$\Delta H_f^\circ[\text{CO}_2(\text{g})] = -395.3 \text{ kJ}$$

$$\Delta H_f^\circ[\text{H}_2\text{O}(\text{l})] = -285.8 \text{ kJ}$$



Use standard enthalpies of formation, available in Appendix A.1 of *Interactive General Chemistry* or below, to determine ΔH° for the following reaction in kilojoules per mole (kJ/mol).



- Some of the amazing things we can now do with thermochemistry...
 1. Define, apply, and carry out calculations related to pressure-volume work.
 2. Define, apply, and carry out calculations related to heat.
 3. Properly define system, surroundings, and universe when solving problems in thermodynamics; classify thermodynamic systems based on matter and energy transfer.
 4. Use the first law of thermodynamics as a basis for energy balances.
 5. Distinguish between state variables and path variables.
 6. Apply the results of a calorimetry experiment to calculate an enthalpy change.
 7. Apply Hess's law to calculate the enthalpy change of an unknown or unmeasured reaction using known enthalpy changes.
 8. Apply standard enthalpies of formation to calculate standard enthalpy of reaction.